

GALACTIC NEIGHBORS

Dark dust lanes are silhouetted against glowing gas and stars in this view of the Andromeda Galaxy and its two close companions, the dwarf elliptical galaxies M32 (upper left) and M110 (bottom).

EXPLORING SPACE**INTERGALACTIC DISTANCE**

The study of M31 played a key role in the discovery that galaxies exist beyond our own. Although the spectra of galaxies suggested they shone with the light of countless stars, no one could measure their immense distance. In 1923, Edwin Hubble (see p.45) proved that M31 lay outside our galaxy. He found the true distance of M31 by calculating the luminosities of its Cepheid variable stars (see pp.282–283) and relating their true brightness to their apparent magnitude.

